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Network nodes, a user equipment and methods therein for handling a connection between the user equipment and a wireless communications network

Abstract

A method performed by a first network node for handling a connection between a wireless device and a wireless communications network. The wireless communications network comprises the first network node. The first network node stores (705) data related to the connection, and further stores an identifier of the wireless device for locating the stored data related to the connection later. The first network node further provides (706) the wireless device with a command to transition from a connected mode into a suspended mode, in which suspended mode the connection is suspended and the wireless device stores the data related to the connection, and further stores the identifier of the wireless device for locating the stored data later in order to resume the connection.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8934336	12/2014	Ramachandran et al.	N/A	N/A
9288434	12/2015	Bangolae	N/A	H04L 65/762
9332461	12/2015	Han et al.	N/A	N/A
2004/0203973	12/2003	Khan	455/517	H04L 47/266
2006/0007862	12/2005	Sayeedi	370/235	H04L 47/10
2007/0097216	12/2006	Koide	348/207.1	H04N 5/232
2008/0263213	12/2007	Kinoshita	709/228	H04L 67/2861
2009/0197575	12/2008	Degraeve	455/412.1	H04L 29/1216
2010/0130233	12/2009	Parker	455/456.3	H04M 1/72538

2010/0315948	12/2009	Yi	370/235	H04L 47/10
2012/0023506	12/2011	Maeckel et al.	N/A	N/A
2012/0173901	12/2011	Soliman	713/320	H04W 88/182
2013/0150126	12/2012	Pattaswamy	455/558	H04W 8/183
2013/0208699	12/2012	Häkkinen et al.	N/A	N/A
2013/0260811	12/2012	Rayavarapu	455/509	H04W 76/19
2013/0297810	12/2012	Ho et al.	N/A	N/A
2014/0022990	12/2013	Guo et al.	N/A	N/A
2014/0071905	12/2013	Nimmala	370/329	H04W 28/18
2014/0080479	12/2013	Vangala et al.	N/A	N/A
2014/0198740	12/2013	Shinojima	370/329	H04L 5/0078
2015/0146685	12/2014	Han et al.	N/A	N/A
2016/0082674	12/2015	Kray	415/209.3	B29C 70/48
2016/0234774	12/2015	Vejlgaard	N/A	H04W 52/0216

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2 557 890	12/2012	EP	N/A
2557889	12/2012	EP	N/A
2557889	12/2012	EP	N/A
2645803	12/2012	EP	N/A
2645803	12/2012	EP	N/A
2645804	12/2012	EP	N/A
2645804	12/2012	EP	N/A
3582580	12/2018	EP	N/A
WO 2012/050841	12/2011	WO	N/A
2013 155212	12/2012	WO	N/A
WO 2014/117854	12/2013	WO	N/A

OTHER PUBLICATIONS

3GPP TR 23.887 v12.0.0; Technical Report; 3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Study on Mechanic-Type Communications (MTC) and other mobile data applications communications enhancements (Release 12), Dec. 2013. cited by applicant

3GPP TS 36.304 v12.2.0; Technical Specification; 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE); procedures in idle mode (Release 12), Sep. 2014. cited by applicant

European Extended Search Report with European Search Opinion in European Application No. 23159431.8-1218, dated Apr. 26, 2023. cited by applicant

PCT Notification of Transmittal of the International Search Report and the Written Opinion of the International Searching Authority, or the Declaration for International application No. PCT/SE2015/051113—Feb. 24, 2016. cited by applicant

3GPP TS 36.304 v12.2.0; Technical Specification; 3rd Generation Partnership Project; Technical

Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) procedures in idle mode (Release 12), Sep. 2014. cited by applicant
3GPP TS 36.331 v12.3.0; Technical Specification; 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification (Release 12), Sep. 2014. cited by applicant

3GPP TS 36.413 v12.3.0; Technical Specification; 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP) (Release 12), Sep. 2014. cited by applicant

3GPP TR 23.890 v12.0.0; Technical Report; 3rd Generation Partnership Project; Technical Specification Group Services and System Aspects; Optimized offloading to Wireless Local Area Network (WLAN) in 3GPP Radio Access Technology (RAT) mobility (Release 12)—Sep. 2013. cited by applicant

PCT Written Opinion of the International Searching Authority for International application No. PCT/SE2015/05111—dated Feb. 15, 2016. cited by applicant

PCT International Search Report for International application No. PCT/SE2015/05111—Feb. 15, 2016. cited by applicant

Supplementary European Search Report for Application No. / Patent No. 15855451.9—1870 / 3213557—Sep. 20, 2017. cited by applicant

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Background/Summary

PRIORITY (1) .[This nonprovisional application is.]. .Iadd.This patent application is for a broadening reissue of U.S. Pat. No. 9,924,460, which is hereby incorporated by reference in its entirety. U.S. Pat. No. 9,924,406 issued from U.S. patent application Ser. No. 14/892,841, .Iaddend.a U.S. National Stage Filing under 35 U.S.C. § 371 of International Patent Application Ser. No. PCT/SE2015/051113 filed Oct. 20, 2015, and entitled “Network Nodes, A User Equipment And Methods Therein For Handling A Connection Between The User Equipment And A Wireless Communications Network” which claims priority to U.S. Provisional Patent Application No. 62/069,382 filed Oct. 28, 2014 and also claims priority to U.S. Provisional Patent Application No. 62/079,821 filed on Nov. 14, 2014, all of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(1) Embodiments herein relate to a first network node, a wireless device, a second network node, a third network node, a system of network nodes and methods therein. In particular the embodiments herein relate to handling a connection between the wireless device and a wireless communications network.

BACKGROUND

(2) Communication devices such as terminals are also known as e.g. User Equipments (UE), mobile terminals, wireless terminals and/or mobile stations. Terminals are enabled to communicate wirelessly in a wireless or cellular communications network or a wireless communication system, sometimes also referred to as a cellular radio system or cellular network. The communication may be performed e.g. between two terminals, between a terminal and a regular telephone and/or between a terminal and a server via a Radio Access Network (RAN) and possibly one or more core

networks, comprised within the cellular communications network.

(3) Examples of wireless communications networks are Long Term Evolution (LTE.Iadd.TM.Iaddend.), Universal Mobile Telecommunications System (UMTS) and Global System for Mobile communications (GSM).

(4) Terminals may further be referred to as mobile telephones, cellular telephones, laptops, surf plates or tablets with wireless capability, just to mention some further examples. The terminals in the present context may be, for example, portable, pocket-storable, hand-held, computer-comprised, or vehicle-mounted mobile devices, enabled to communicate voice and/or data, via the RAN, with another entity, such as another terminal or a server.

(5) The cellular communications network covers a geographical area which is divided into cell areas, wherein each cell area being served by an access node such as a base station, e.g. a Radio Base Station (RBS), which sometimes may be referred to as e.g. “eNB”, “eNodeB”, “NodeB”, “B node”, or BTS (Base Transceiver Station), depending on the technology and terminology used. The base stations may be of different classes such as e.g. macro eNodeB, home eNodeB or pico base station, based on transmission power and thereby also cell size. A cell is the geographical area where radio coverage is provided by the base station at a base station site. One base station, situated on the base station site, may serve one or several cells. Further, each base station may support one or several communication technologies. The base stations communicate over the air interface operating on radio frequencies with the terminals within range of the base stations. In the context of this disclosure, the expression Downlink (DL) is used to denote the transmission path from the base station to the mobile station. The expression Uplink (UL) is used to denote the transmission path in the opposite direction i.e. from the mobile station to the base station.

(6) In 3rd Generation Partnership Project (3GPP) LTE, base stations, which may be referred to as eNodeBs or even eNBs, may be directly connected to one or more core networks and related core network nodes, such as a Mobility Management Entity (MME) and a Serving GateWay (SGW).

(7) The MME is a control node and is responsible for idle mode UE tracking and paging procedure including retransmissions. The MME is involved in the bearer activation/deactivation process and is also responsible for choosing the SGW for a UE at the initial attach and at time of intra-LTE handover involving Core Network (CN) node relocation. The MME is responsible for authenticating the user.

(8) 3GPP LTE radio access standard has been written in order to support high bitrates and low latency both for uplink and downlink traffic. All data transmission is in LTE controlled by the radio base station.

(9) The introduction of new types of devices, such as devices that are used for Machine-Type-Communication (MTC), that interact with wireless communications networks put new requirements both on the devices, such as UEs, and the networks. Such new requirements may impose problems, such as shorter battery lifetime for the UEs, and from the network point of view one of the problems is to handle many devices sending small amounts of data.

(10) For example, when a device, such as a UE, wants to send data, it needs to set up a connection towards the wireless communications network. This implies some mandatory steps, e.g. establishing a control connection to the MME through the eNodeB, establishing a secure radio connection on the air interface and configuring data bearers through which data may be sent. FIG. 1 shows a legacy normal service request establishment in LTE when data is to be received or transferred. As can be seen from FIG. 1 there is rather much signaling, i.e. actions **101-114**, before the UE sends **115** the payload data to the eNodeB.

(11) The signaling described above consumes much processing resources compared to the data volume being transferred for some devices, such as MTCs. This may then set a limit on how many devices that an eNodeB may handle, or how much other traffic the eNodeB may handle.

(12) One prior art approach is to always have the devices in connected mode, that is e.g. have a control connection towards the MME, a secure radio connection and configured data bearers

established all the time. This will however have impacts both on the performance of mobile devices, e.g. in terms of battery lifetime, and also on the performance of the network, since the connected mode consumes network resources. In fact, an eNodeB is normally designed to only handle a certain maximum number of users in connected mode. Since the actual number of devices in the area may be multiples higher than the designed maximum number of users in connected mode this will impose a large limitation.

SUMMARY

(13) It is an object of embodiments herein to address at least some of the problems and issues mentioned above, by providing an improved way of handling a connection between a wireless device and a wireless communications network.

(14) When a connection between the UE, i.e. wireless device, and the network is released in LTE, all contexts in the eNodeB and the related information in the MME and the SGW are released. At the next service request, e.g. when the UE wants to send data, the entire signaling procedure for establishing a connection needs to be performed again. The current way to always establish a new connection leads to much signaling and hence consume much processing resources compared to the data volume being transferred for some devices, such as MTCs.

(15) Instead of releasing an established connection between a wireless device, such as the UE, and the wireless communications network, the wireless device is 'suspended' in embodiments herein. Suspended means that data related to the established connection between the wireless device and the wireless communications network are saved in the wireless device and in network nodes connected to the wireless device, such as an eNodeB, an MME and an SGW.

(16) The data may comprise a UE context, for example a UE context related to security, bearers, Temporary International Mobile Subscriber Identity (T-IMSI) and UE capabilities.

(17) Further, in embodiments herein, instead of establishing a new connection between the wireless device and the wireless communications network when there is new data to be transferred, the previous connection is resumed, or continued, at a later time with the same security context and bearers.

(18) In other words, in embodiments herein the wireless device may be connected to the wireless communications network using the same connection parameters for several sessions of the connection since these connection parameters are saved in the wireless device and in the network nodes during the above suspension of the connection.

(19) The wireless device is in other aspects handled as in 'Idle mode'. Although the wireless device is in other aspects handled as in 'Idle mode', the wireless device may continue a previous session of the connection, i.e. 'resume the previous session' when the wireless device wants to establish a connection towards the network, e.g. when there is new data to be transferred.

(20) The 'Resume' of the previous connection may be allowed within the cell where the wireless device was 'suspended' or in a cell in which information indicating that resume is possible is broadcasted.

(21) According to one aspect of embodiments herein, the object is achieved by a method performed by a first network node for handling a connection between a wireless device and a wireless communications network. The wireless communications network comprises the first network node.

(22) The first network node stores data related to the connection, and further stores an identifier of the wireless device for locating the stored data related to the connection later.

(23) The first network node further provides the wireless device with a command to transition from a connected mode into a suspended mode. In the suspended mode the connection is suspended and the wireless device stores the data related to the connection, and further stores the identifier of the wireless device for locating the stored data later when resuming the connection.

(24) According to another aspect of embodiments herein, the object is achieved by a method performed by a wireless device for handling a connection between the wireless device and a wireless communications network. The wireless communications network comprises the first

network node.

(25) The wireless device receives from the first network node a command to transition from a connected mode into a suspended mode.

(26) The wireless device further transitions into the suspended mode. In the suspended mode the connection is suspended and the wireless device stores data related to the connection, and further stores an identifier of the wireless device for locating the stored data related to the connection later when resuming the connection.

(27) According to a further aspect of embodiments herein, the object is achieved by a method performed by a second network node for handling a connection between a wireless device and a wireless communications network. The wireless communications network comprises the second network node and a third network node. The second network node is responsible for paging the wireless device.

(28) The second network node obtains an indication that the connection between the wireless device and the wireless communications network is to be suspended.

(29) The second network node further pages the wireless device in case of a downlink control message is to be transferred to the wireless device or when the third network node has informed the second network node that a payload is to be transferred to the wireless device.

(30) According to a further aspect of embodiments herein, the object is achieved by a method performed by a third network node for handling a connection between a wireless device and a wireless communications network. The wireless communications network comprises the third network node and a second network node responsible for paging of the wireless device.

(31) The third network node obtains an indication that the connection between the wireless device and the wireless communications network is to be suspended.

(32) The third network node further suspends a bearer related to the wireless device in response to the indication that the connection between the wireless device and the wireless communications network is to be suspended. Suspending the bearer comprises buffering downlink payload to be sent to the wireless device using the bearer.

(33) When the third network node has received downlink payload to the wireless device for which the bearer is suspended, the third network node informs the second network node that the third network node has received downlink payload to the wireless device.

(34) According to a further aspect of embodiments herein, the object is achieved by a method performed by a system of network nodes for handling a connection between a wireless device and a wireless communications network. The wireless communications network comprises the system of network nodes. A first network node comprised in the system of network nodes stores data related to the connection, and further stores an identifier of the wireless device for locating the stored data related to the connection later.

(35) The first network node provides the wireless device with a command to transition from a connected mode into a suspended mode. In the suspended mode the connection is suspended and the wireless device stores the data related to the connection, and further stores the identifier of the wireless device for locating the stored data later when resuming the connection.

(36) A second network node comprised in the system of network nodes receives a message from the wireless device. The message indicates a request from the wireless device to resume the connection. The received message comprises the identifier of the wireless device for locating the stored data. The identifier enables the first network node to locate the data related to the connection.

(37) The second network node resumes the connection by transmitting a message to the wireless device. The transmitted message comprises the identifier of the wireless device, and further comprises an indication that a resumption of the connection is complete, whereby the suspended connection between the wireless device and the wireless communications network is resumed using the data related to the suspended connection.

- (38) According to a further aspect of embodiments herein, the object is achieved by a first network node configured to perform the above method performed by the first network node.
- (39) According to a further aspect of embodiments herein, the object is achieved by a wireless device configured to perform the above method performed by the wireless device.
- (40) According to a further aspect of embodiments herein, the object is achieved by a second network node configured to perform the above method performed by the second network node.
- (41) According to a further aspect of embodiments herein, the object is achieved by a third network node configured to perform the above method performed by the third network node.
- (42) According to a further aspect of embodiments herein, the object is achieved by a system of network nodes configured to perform the above method performed by the system of network nodes.
- (43) Since the first network node commands the wireless device to transition into a suspended idle mode, also referred to as a suspended mode herein, in which suspended idle mode the wireless device has stored the data related to the connection, and the identifier of the wireless device for locating the data related to the connection, the wireless device and the wireless communications network are able to resume the connection later.
- (44) In other words, since the data related to the connection and the identifier of the wireless device for locating the data related to the connection is stored in the wireless device and in the wireless communications network it is possible to suspend and resume the connection using the data related to the connection.
- (45) Embodiments herein enables a faster transition from 'idle' to 'ready to transfer data mode', such as a Radio Resource Control, RRC, Connected Mode. In embodiments herein no round trip communication towards the core network node is needed to set up a connection towards the wireless communications network. This will significantly reduce the time before an uplink packet may be transferred. Note also that the reduced time before an uplink packet may be transferred will be the same when the S1 connection is over satellite, i.e. 20-30 ms compared to 1 s with the current method where a new connection is needed every time new data is to be sent from the UE to the network. Further, when the S1 connection is over satellite, also the time to transfer a DL package from the wireless communications network to the UE is significantly improved, 250 ms compared to 1 s for the current method.
- (46) Embodiments herein require less signaling and hence consume less resources on the air interface, in the eNodeB, on an S1-C interface between the eNodeB and the MME, in the MME, on the S11 interface between the MME and the SGW and in the SGW.
- (47) By using at least some of the embodiments described herein, considerable advantages may be achieved e.g. for UEs with low mobility which transfer rather small amounts of data at a time. However, benefits can be achieved also in other cases which will be further described below. For example, normal smart phone traffic will gain faster setup times and the underlying wireless communications network will need less processing.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Examples of embodiments herein are described in more detail with reference to attached drawings in which:
- (2) FIG. 1 is a signaling diagram illustrating a method for accessing the wireless communications network according to prior art.
- (3) FIG. 2a is a schematic block diagram illustrating a wireless communications network in which embodiments herein may be implemented.
- (4) FIG. 2b is a further schematic block diagram illustrating a wireless communications network in which embodiments herein may be implemented.

- (5) FIG. 2c is a schematic block diagram illustrating data related to a connection and an identifier of a wireless device used for handling the connection.
- (6) FIG. 3 is a signaling diagram illustrating embodiments of a method performed in a wireless communications network.
- (7) FIG. 4a is a signaling diagram illustrating further embodiments of a method performed in a wireless communications network.
- (8) FIG. 4b is a signaling diagram illustrating further embodiments of a method performed in a wireless communications network.
- (9) FIG. 4c is a signaling diagram illustrating further embodiments of a method performed in a wireless communications network.
- (10) FIG. 4d is a signaling diagram illustrating further embodiments of a method performed in a wireless communications network.
- (11) FIG. 5 is a flowchart illustrating embodiments of a method performed by a first network node.
- (12) FIG. 6 is a flowchart illustrating embodiments of a method performed in a wireless communications network.
- (13) FIG. 7 is a flowchart illustrating embodiments of a method performed by a first network node.
- (14) FIG. 8 is a flowchart illustrating embodiments of a method performed by a wireless device.
- (15) FIG. 9 is a flowchart illustrating embodiments of a method performed by a second network node.
- (16) FIG. 9a is a flowchart illustrating further embodiments of a method performed by a second network node.
- (17) FIG. 10 is a flowchart illustrating embodiments of a method performed by a third network node.
- (18) FIG. 10a is a flowchart illustrating further embodiments of a method performed by a third network node.
- (19) FIG. 11 is a schematic block diagram illustrating embodiments of a first network node.
- (20) FIG. 12 is a schematic block diagram illustrating embodiments of a wireless device.
- (21) FIG. 13 is a schematic block diagram illustrating embodiments of a second network node.
- (22) FIG. 14 is a schematic block diagram illustrating embodiments of a third network node.

DETAILED DESCRIPTION

- (23) Embodiments herein may be implemented in one or more wireless communications networks whereof FIG. 2a depicts parts of a wireless communications network **200**, according to one example. The wireless communications network **200** may for example be an LTE, UMTS, GSM, or any 3GPP wireless communications network, or any cellular wireless communications network or system.
- (24) The wireless communications network **200** comprises a plurality of base stations and/or other network nodes. More specifically, the wireless communications network **200** comprises a first network node **211**.
- (25) The term “network node” may correspond to any type of radio network node or any network node, which communicates with at least a radio network node. For example, the network node **211** may be a base station, such as an eNB. The base station may also be referred to as a NodeB, an evolved Node B (eNB, eNode B), a base transceiver station (BTS), Access Point (AP) Base Station, Wi-Fi AP, base station router, or any other network unit capable of communicating with a wireless device within a cell served by the base station depending e.g. on the radio access technology and terminology used. The network node **211** may also be a Radio Network Controller (RNC) in an UMTS system. In this description, the equivalent term user equipment, UE, will sometimes be used instead of wireless device and these terms are interchangeable throughout the description.
- (26) The wireless communications network **200** may further comprise a second network node **212**, for example a first core network node such as an MME. The second network node **212** may be a control node and may be responsible for idle mode UE tracking and paging procedure including

retransmissions. The second network node **212** may be involved in the bearer activation/deactivation process and may also be responsible for choosing an SGW for a UE at the initial attach and at time of intra-LTE handover involving CN node relocation. The second network node **212** may be responsible for authenticating the user. Embodiments herein are applicable for “MME in Pool”, wherein the eNodeB may be connected to a number of MME nodes serving the same cells, or for similar technical solutions.

(27) The wireless communications network **200** may further comprise a third network node **213**, for example a third network node such as an SGW.

(28) In some embodiments, which may be described with reference to FIG. 2b, the wireless communications network **200** further comprises a fourth network node **214**, e.g. another base station such as another eNB or an RNC in an UMTS system. The first network node **211** and the fourth network node **214** may operate and/or be comprised in a system **216** of network nodes depicted in FIG. 2b. Some of the embodiments herein may be implemented in the system **216** of network nodes. For example, a first part of these embodiments may be performed in the first network node **211** while a second part of these embodiments may be performed in the fourth network node **214**.

(29) The network node **211** serves a cell **221**. Likewise, the fourth network node **214** may serve one or more cells. In FIG. 2b the fourth network node **214** serves a second cell **224**.

(30) A cell is a geographical area where radio coverage is provided by network node equipment such as Wi-Fi AP equipment, base station equipment at a base station site or at remote locations in Remote Radio Units (RRU). The network node **211** is an example of such network node equipment. The cell definition may also incorporate frequency bands and radio access technology used for transmissions, which means that two different cells may cover the same geographical area but using different frequency bands. Each cell is identified by an identity within the local radio area, which is broadcast in the cell. Another identity identifying cells uniquely in the whole of a wireless communication network is also broadcasted in the cells.

(31) Throughout this description, the term “cell” is used to represent any area in which a network node provides radio coverage for wireless devices. Hence, this description is not limited to cellular networks and it may be valid and useful for any type of radio network in which various network nodes provide radio coverage.

(32) Network nodes, such as base stations and Wi-Fi AP, communicate over the air or radio interface operating on radio frequencies with user equipment within range of the network nodes. The user equipment transmit data over the radio interface to network nodes, such base stations and Wi-Fi AP, in UL transmissions, and network nodes, such as Wi-Fi AP and base stations, transmit data over an air or radio interface to the user equipment in DL transmissions.

(33) The first network node **211** communicates with wireless devices in the cell **221**, such as a wireless device **240**, also referred to as a UE or a user equipment. The wireless device **240** may be in different states or modes of operation. For example, when no data is sent from or to the wireless device **240** it may be in an idle state, in which the wireless device **240** does not have a connection **251**, such as a Radio Resource Control (RRC) connection, in place with the first network node **211**, such as an eNB. When data is transmitted to or from the wireless device **240** it may be in an active state with an active connection **251** between the wireless device **240** and the wireless communications network **200**.

(34) The connection **251** may be described by data **260** related to the connection **251**. The data **260** related to the connection **251** is illustrated in FIG. 2c. The data **260** related to the connection **251** may be a context, such as a UE context. The UE context is related to the connection **251**. As mentioned above, the UE context may comprise data related to the connection **251**, such as data **262** related to security, such as keys, sequence number etc, and data **263** related to bearers, such as signaling and data bearers. The connection **251** between the wireless device **240** and the wireless communications network **200** comprises one or more bearers which are “virtual” connections

between two endpoints in the wireless communications network **200**. For example, in LTE signaling bearers are virtual connections between a UE and an MME, while data bearers are virtual connections between the UE and the SGW.

(35) The wireless device **240** may be an MTC user equipment.

(36) The wireless device **240** may further be e.g. a mobile terminal or a wireless terminal, a mobile phone, a computer such as e.g. a laptop, a Personal Digital Assistants (PDAs) or a tablet computer, sometimes referred to as a surf plate, with wireless capability, target device, device to device UE, MTC UE or UE capable of machine to machine communication, iPad, mobile terminals, smart phone, Laptop Embedded Equipment (LEE), Laptop Mounted Equipment (LME), USB dongles etc. or any other radio network units capable to communicate over a radio link in a wireless communications network. Please note the term user equipment used in this document also covers other wireless devices such as Machine to machine (M2M) devices, even though they do not have any user.

(37) Embodiments herein provide a more efficient procedure to move the wireless device **240**, such as a UE, from a state corresponding to an idle state, e.g. RRC_IDLE state or mode, to a state where data may be transferred to and from the wireless communications network **200**. This is achieved by reusing information from the previous session, i.e. from the previous connection.

(38) The procedure is very efficient for wireless devices that make many accesses within the same cell or cell group. In this case, both air interface signaling as well as network internal signaling is reduced.

(39) Embodiments herein will now be described briefly in general terms to give the reader an overview of embodiments herein.

(40) Instead of releasing an established connection **251** between the wireless device **240** and the wireless communications network **200**, the wireless device **240** and the connection **251** are 'suspended' in embodiments herein. Suspended means that any data **260** related to the established connection **251** between the wireless device **240** and the wireless communications network **200**, including connection parameters, is saved in the wireless device **240** and in network nodes **211**, **212**, **213** connected to the wireless device **240**, such as an eNodeB, an MME and an SGW.

(41) The data **260** related to the connection **251** may comprise a UE context, for example a UE context related to security, bearers, T-IMSI and UE capabilities. More specifically, the UE context may comprise a security context and a context for the bearer, both related to the previous connection **251**.

(42) Further the wireless device **240** and the first network node **211** save the identifier **270** of the wireless device **240** for locating the data **260** related to the connection **251** later.

(43) The wireless device **240** is in other aspects handled as if being in 'Idle mode', but when wanting to establish a connection towards the wireless communications network, the wireless device **240** may continue a previous session, that is 'resume the previous session' of the already established connection. I.e. in embodiments herein the wireless device **240** may be connected to the wireless communications network **200** using the same connection parameters for several sessions of the connection since these connection parameters are saved in the wireless device **240** and in the network nodes **211**, **212**, **213** during the above suspension of the connection **251**.

(44) Further, in embodiments herein, instead of establishing a new connection between the wireless device **240** and the wireless communications network **200** when there is new data to be transferred, the previous connection **251** is resumed, or continued, at a later time using the same security context and bearers.

(45) The previous connection **251** is resumed by using the identifier **270** of the wireless device **240** for locating the data **260** related to the connection **251**.

(46) The 'Resume' of the previous connection **251** may be allowed within the cell **221** where the wireless device was 'suspended' or in the cell **221** in which information indicating that resume is possible is broadcasted.

(47) Below, methods for suspending and resuming a connection between the wireless device **240** and the wireless communications network **200** will be illustrated in more detail by means of exemplary embodiments. The following embodiments will be described using LTE as an example, although the solution is not limited thereto, and the first network node **211** will be an LTE base station, i.e. an eNB. Further the second network node **212** will be an MME and the third network node **213** will be an SGW.

(48) It will be assumed that network data is kept in the eNodeB, the MME and the SGW and that the S1-bearers are kept. Network data may for example be data related to the UE context and to the setup of bearers in the different network nodes. For example, when the first network node **211** is an eNB the network data may comprise MME resource used, SGW S1-U addresses, bearer configuration and Security context information. When the second network node **212** is an MME the network data may comprise UE and subscriber information, e.g. comprising bearer configuration, PDN connection used, SGW resource used and eNB resource used. When the third network node **213** is an SGW the network data may comprise PDN connection used, eNB S1-U addresses and Bearer configuration. Further, the S1 setup remains during the suspension of the connection. The radio bearers are remembered in the UE, eNB and MME and ‘restarted’ again at resume.

(49) It should be noted that these embodiments are not mutually exclusive. Components from one embodiment may be tacitly assumed to be present in another embodiment and it will be obvious to a person skilled in the art how those components may be used in the other exemplary embodiments.

(50) Actions that may be performed for handling the connection **251** between the wireless device **240** and the wireless communications network **200** according to embodiments herein will now be described in more detail with reference to FIG. 3, FIG. 4a, FIG. 4b, FIG. 4c and FIG. 4d. FIG. 3 and FIGS. 4a to 4d show the interaction between the different nodes that take part in embodiments herein.

(51) FIG. 3 is a signaling diagram illustrating a communication between the wireless device **240**, the first network node **211**, the second network node **212** and the third network node **213**. FIG. 3 describes actions that may take place to resume the connection **251** between the wireless device **240**, such as a UE, and the wireless communications network **200** according to embodiments herein. Further, FIG. 3 shows the interactions between the wireless device **240**, the first network node **211**, the second network node **212** and the third network node **213**.

(52) More specifically, FIG. 3 describes actions that may be performed to suspend the connection between the wireless device **240** and the wireless communications network **200**.

(53) FIG. 4a is a further signaling diagram that describes actions **401-410** that may take place to resume the connection **251** between the wireless device **240** and the wireless communications network **200** according to embodiments herein. More specifically, FIG. 4a describes further actions that may be performed to resume the connection **251** between the wireless device **240** and the wireless communications network **200** from its suspended state. FIGS. 4b, 4c and 4d disclose optional details.

(54) First the connection between the wireless device **240** and the wireless communications network **200** is established in a conventional manner as described by actions **301-314** in FIG. 3. UL and DL data is transmitted in action **315** and **316**. Thus the wireless device is in a connected mode, e.g. RRC Connected mode, as illustrated with the state/mode **500** in FIG. 5. The data bearers are set up. At some point in time the wireless device **240** has indicated that it supports the ‘suspend/resume’ handling. The support of suspend/resume may be implemented as a capability together with other UE capabilities. The capability may be sent to the wireless communications network **200** when setting up a connection, such as the connection **251**, unless the wireless communications network **200** already has the capability.

(55) After some time there is a timeout related to the connection. The timeout may be a result of a timer that expires in the first network node **211**. The timer is related to the connection. However, instead of releasing the connection **251** it is ‘suspended’, which is described in actions **317** to **321**,

which will be described in more detail below. The first network node **211** may for example decide to suspend the connection **251** if the wireless device **240** supports suspension.

(56) The concerned core network nodes, such as the second network node **212** and the third network node **213** are informed, for example by the first network node **211**, that the wireless device **240** is in an 'RRC Suspended state'. The second network node **212** and the SGW may be informed either directly by the first network node **211** or via the other core network node. For example, the SGW may be informed by the second network node **212**. In FIG. 3 the second network node **212** is informed about the wireless device **240** being 'suspended', for example with S1-AP UE Ctxt Deactive in action **317**. The SGW is signaled with Suspend Bearer in action **319**, to stop DL data traffic by the first network node **211**.

(57) The second network node **212** may acknowledge the information from the first network node **211** in action **318** and the third network node **213** may acknowledge the information from the first network node **211** in action **321**.

(58) The wireless device **240** receives **320** a command, e.g. from the first network node **211**, to transition into a suspended idle mode. The wireless device **240** will then behave as in 'idle mode', but caches a context related to the last connection, for example data related to security, such as keys, sequence number etc, and data related to bearers, such as signaling and data bearers.

(59) The bearers, such as signaling and data bearers, related to the previous connection are kept, i.e. these bearers are maintained for the time being by the network on behalf of the wireless device **240**.

(60) The wireless device **240** is further given a unique identity or identifier **270** of the wireless device **240**, e.g. a resume identifier, to use later when the wireless device **240** wants to resume the connection. The identifier **270** of the wireless device **240**, such as the resume identifier, is used to locate the data **260** related to the connection **251**, such as the UE context, within the first network node **211**. The C-RNTI (MAC id) may be used as the identifier **270**. The identifier **270**, such as the resume identifier, is unique within an area, which comprises one or more cells, such as the cell **221**. If the area is more than one cell, the cells may broadcast information to allow the wireless device **240** to decide if resumption is allowed or not in the cells.

(61) Also other information may be broadcasted to allow the wireless device **240** to decide if resumption is allowed in the cells, e.g. after how long time resumption is allowed in the cells, validity of the resume identifier etc.

(62) The wireless device **240** is now in a suspended state. The suspended state corresponds to an idle state in which the wireless device **240** stores the data **260** related to the connection **251** and the identifier **270** of the wireless device **240** for resuming the connection **251** later.

(63) When the wireless device **240** wants to resume its connection **251** for some reason, e.g. due to paging of the wireless device **240**, wanting to send data, etc, it takes contact with the network, i.e. with the first network node **211**. If the wireless device **240** supports a resumption of the suspended connection **251** it may check whether or not the identifier **270** of the wireless device **240** is valid.

(64) The initial contact may be made by transmitting **401** a first random access message RA msg 1 comprising a preamble. The first network node **211** may respond **402** with a second random access message msg2. Then the wireless device **240** initiates the resumption by transmitting **403a** either a new message RA msg3, called RRC Connection Resume Request in embodiments herein, or an updated existing message, e.g. RRC Re-establish Request. The message comprises data which enable the first network node **211** to uniquely identify the previously suspended wireless device **240**. Such data may for example be the identifier **270** of the wireless device **240** which is related to the connection.

(65) The RA msg 3 message may also comprise a kind of 'authentication token' to reduce the risk of 'Denial of service' attacks and also fraud.

(66) If the first network node **211** finds that the identifier **270** of the wireless device **240**, such as the resume identifier, sent to it from the wireless device **240** in order to resume the connection **251**, is known and the data **260** related to the connection **251**, e.g. the related UE context, is ok, e.g. the

stored UE context has not been deleted or become too old, then the first network node **211** responds to the wireless device **240** with a response message in action **405**. The response message may for example be a control message such as RRC Connection Resume Pending or RRC Connection Resume Complete. The response message comprises the identifier **270** of the wireless device **240**, such as the resume identifier, to make sure both the wireless device **240** and the first network node **211** communicates about the same connection. This may be needed e.g. for performing contention resolution, and informs the wireless device **240** that the resumption is complete.

(67) The response message may further indicate to the wireless device **240** that the first network node **211** recognises the wireless device **240** on RRC level, enable contention resolution and give the wireless device **240** a unique scheduling identifier, e.g. C-RNTI in LTE.

(68) To allow for different response times for contention resolution—the objective is normally to keep the response time short—and for final completion of the resumption the response message may be split into two messages or kept as one message. That is, as illustrated in FIG. **4a**, a first response message, RRC Connection Pending, is sent in action **404** to complete the contention resolution by giving a unique scheduling identifier within a certain short time. Then, afterwards the first network node **211** finally accepts the resumption with the later RRC Connection Complete message in action **405**. The first network node **211** may also chose to only send RRC Connection Complete and include all data needed for the contention resolution and the final completion of the resumption of the connection.

(69) When this has been performed the wireless device **240** may be scheduled by the first network node **211** to send uplink data on the resumed bearers. The action of scheduling is not shown in FIG. **4a**.

(70) The first network node **211** then informs **406** the second network node **212** that the data **260** related to the connection **251**, e.g. the UE context, has become active, that is, paging is not needed anymore. The first network node **211** further informs **407** the third network node **213** that DL packets may be sent on the DL data bearers. The actual signaling may be made in many ways. In FIG. **4a** the first network node **211** signals the second network node **212** with an S1-AP UE Ctxt Active and sends in-band instructions, such as Bearer resume to the third network node **213**. It is also possible that the second network node **212** informs the third network node **213** instead or vice versa.

(71) The wireless device **240** may now send uplink data to the third network node **213** in action **408**.

(72) The second network node **212** responds with an S1-AP UE Ctxt Active Ack to the first network node **211** in action **409**. The third network node **213** responds with a Bearer Resume Ack in action **410**.

(73) When the third network node **213** has been informed that the data **260** related to the connection **251**, e.g. the UE context, has become active, it may forward **411** DL data or messages to the first network node **211** to be transferred to the wireless device **240**.

(74) The wireless device **240** is now fully resumed and in the same state in all nodes as if a normal service establishment had been done.

(75) When a network node decides that the wireless device **240** shall not be kept in RRC_CONNECTED anymore, this may be either initiated by the first network node **211** or initiated by the second network node **212**.

(76) FIG. **4a** illustrates the case where it is the first network node **211** that initiates to stop the connected mode. The first network node **211** then sends **412** a request to the second network node **212** to deactivate the data **260** related to the connection **251**, e.g. the UE context. In other words, the first network node **211** requests that the second network node **212** pages the wireless device **240** in case of downlink signaling or data is to be sent to the wireless device **240**. The second network node **212** answers **413** if this was accepted, e.g. if the second network node **212** supports suspended mode functionality and suspended mode is allowed to use. In the case of acceptance, the first

network node **211** will inform **414** the third network node **213** to suspend DL packet transfer on DL data bearers and to inform the second network node **212** to page the wireless device **240** in case of a downlink control message or a payload, i.e. data traffic, is to be transferred to the wireless device **240**. The first network node **211** will also command **415** the wireless device **240** to transition into suspended idle mode. The command comprises the identifier **270** of the wireless device **240**, e.g. a resume id, to be used at a possible later resumption of the connection.

(77) In the case that it is the second network node **212** that initiates to stop the connected mode, the second network node **212** may send a UE Context Release Command to the first network node **211**. The command may comprise an optional indication that the connection **251** may be suspended as an alternative to released. If the first network node **211** decides to suspend the connection instead of releasing the connection, it should indicate this in a response message to the second network node **212**. The first network node **211** will then continue to indicate to the third network node **213** to suspend DL packet transfer on DL bearers. The first network node **211** will also indicate to the second network node **212** to page the wireless device **240**. The first network node **211** will also command the wireless device **240** to be suspended, i.e. transition into suspended idle mode. The command comprises the identifier **270** of the wireless device **240**, e.g. the resume identifier, to be used at a possible later resumption.

(78) If a DL packet is sent from the third network node **213** to the first network node **211** on a bearer that the first network node **211** considers suspended, the first network node **211** may send back an error message indicating this. This mechanism will make it possible to not send an acknowledgement on the suspend message from the first network node **211** to the third network node **213** in action **321**, **416**. That is, the acknowledgement of the Suspend Bearer message, i.e. the Bearer Suspended message, in FIG. 3 and FIG. 4a may in that case be removed.

(79) Also the acknowledge message in action **410** for the Bearer Resume message in action **407** from the first network node **211** to the third network node **213** may be omitted if the first network node **211** repeatedly re-sends the 'Bearer Resume' message until a DL message has been detected.

(80) FIG. 4b describes how the data bearers may be kept suspended also after resumption of the connection **251**. In this case, the data bearers may be kept suspended in the core network, i.e. in the second and third network nodes **212**, **213**. The data bearers are still known to the wireless device **240**, the first network node **211** and to the core network nodes, i.e. the second network node **212** and the third network node **213**, but may not be used. Only default signaling bearers may be used.

(81) Initially, FIG. 4b corresponds to FIG. 4a, i.e. actions **401b** to **406b** correspond to actions **401a** to **406a**. Action **407c** corresponds to action **409a**. However, the RA msg 3 message in action **403b** may also comprise an indication if the data bearers from the previous connection **251** shall be used or not. Correspondingly, the S1-AP UE Ctxt Active message in action **406b** comprises an indication to the second network node **212** that the bearers for the connection **251** shall be kept suspended.

(82) This allows further advantages for 'signaling only traffic', e.g. Non-Access Stratum (NAS) traffic as exemplified in actions **408b** NAS message UL and **409b** NAS message DL. That is, if only control signaling is needed, e.g. network internal signaling or SMS, then no data bearers are needed.

(83) The connection **251** may be suspended again after the NAS traffic has ended as exemplified with actions **410b-412b** which correspond to actions **412a**, **413a** and **415a**.

(84) The advantages of keeping the data bearers suspended may for example be less processing in the third network node **213** and the first network node **211**. Actions corresponding to actions **407a** and **410a** in FIG. 4a are not performed in FIG. 4c.

(85) If the wireless device **240** or the core network nodes **212**, **213** want to send data, the data bearers may be 'resumed' from the first network node **211** afterwards as described in FIG. 4c. That is, the bearers may be resumed after the connection **251** has been resumed with suspended bearers as described in FIG. 4b. Actions **401c** to **409c** correspond to actions **401b** to **409b**.

(86) Towards the wireless device **240** either a new or existing message, such as an updated RRC

Re-configuration Request in action **410c**, may be reused for later resumption of the data bearers. That is, the wireless device **240** may request the bearers related to the connection **251** to be resumed by sending the updated RRC Re-configuration Request in action **410c** to the first network node **211**. The first network node **211** then sends an S1-AP UE Ctxt Active message to the second network node **212** in action **411c**. The S1-AP UE Ctxt Active message indicates that the bearers shall be active. The first network node **211** sends an RRC Connection Reconfiguration Complete message to the wireless device **240** in action **413c**. The wireless device **240** may now send uplink data to the third network node **213** in action **414c**.

(87) Further, the second network node **212** responds to the first network node **211** with a S1-AP UE Ctxt Active Ack message in action **415c**, as was also the case for action **409** and action **407b**.

(88) Towards the third network node **213** the same signaling as used for the initial resumption of the connection **251** may be used for this later resumption of the data bearers, i.e. action **412c** corresponds to action **407** Bearer resume while action **416c** corresponds to action **410** Bearer Resume Ack.

(89) Actions **417c-419c** suspends the connection **251** again and these actions correspond to actions **410b-412b** in FIG. 4b and actions **412-413** and **415** in FIG. 4a.

(90) When the wireless device **240** is paged, the normal case is to always resume the data bearers. An alternative to this is if the second network node **212** indicates in a paging message if data bearers are needed to be resumed or not. The wireless device **240** may then indicate the need for data bearers to be resumed or not in the following RRC Connection Resume Request.

(91) The first network node **211** may provide additional information to the wireless device **240** in its cells, such as the cell **221**. The first network node **211** may indicate if resume is allowed in the cell **221**. The first network node **211** may indicate which 'Resume Area' the cell **221** belongs to.

(92) The second network node **212** may provide the first network node **211** with information if 'suspend is allowed' during the service establishment and handover if second network node **212** is changed.

(93) It may also be possible to forward the information related to the connection **251** and stored in one network node between the nodes, e.g. over X2 or S1 interfaces to support the resume procedure over a larger geographical area. For example, the wireless device **240** may be suspended by the first network node **211** which stores the data **260** related to the connection **251**. Then the wireless device **240** may resume the connection **251** in a cell served by another network node, e.g. in the second cell **224** served by the fourth network node **214**. The first network node **211** may e.g. forward the data **260** related to the connection **251** to the fourth network node **214**.

(94) In some embodiments the data **260** is stored in the second network node **212** or found via the second network node **212**. In order for the wireless device **240** to be able to initiate locating of the data **260** which is stored in the second network node **212** or in the fourth network node **214** the data **260** may comprise information about the identity of the second network node **212**. For example, the wireless device **240** may store information about the identity of which MME to contact.

(95) FIG. 4d describes some optional actions that may be performed in case the wireless device **240** is suspended by one network node and resumed by another network node. Actions **401d-405d** are similar to actions **401-405**.

(96) In action **406d** the fourth network node **214** signals the second network node **212** to obtain the data **260**. If the data **260** is obtained from another network node, such as the first network node **211**, a path switch is made.

(97) The fourth network node **214** sends **408d** a path switch to the second network node **212** which will signal **409** the third network node **213** to change user plane address, e.g. eNB user plane address, from the first network node **211** to the fourth network node **214**. Then these signals are returned back in action **410d** and action **411d**. I.e. the third network node **213** responds to the second network node **212** in action **410d** and the second network node **212** responds to the fourth network node **214** in action **411d**. The first network node **211** releases the data **260**, e.g. after

receiving **412d** a UE Context Release message from the fourth network node **214**. Then the fourth network node **214** sends **413d** resume complete to the wireless device **240**.

(98) In FIG. **4d** the data **260** is fetched in another network node via the second network node **212**. However, the data **260** may also be obtained directly from the other network node.

(99) There is no need for extra mobility signaling for the wireless device **240**, such as a UE, in suspended idle mode. One or a combination of the following mechanisms may be used to delete the data **260** related to the connection **251**, e.g. the UE context, and related resources in the network nodes, such as the first network node **211**, the second network node **212** and the third network node **213**. An example of related resources are the reserved SGW resources.

(100) If the wireless device **240** makes a service request and the second network node **212** detects that this wireless device **240** is 'suspended', the second network node **212** may inform both the first network node **211** and the third network node **213** to release its resources and contexts related to the suspended data **260** related to the connection **251**, e.g. the suspended UE context.

(101) If the second network node **212**, the first network node **211** or the third network node **213** have deleted, wants to or needs to delete a stored data **260** related to the connection **251**, such as the UE context, then it informs the other network nodes. This may also be done indirectly, e.g. the first network node **211** may only inform the second network node **212** since according to normal legacy logic the second network node **212** releases the third network node **213** resources.

(102) A trigger for such actions may be:

(103) Resource shortage such as memory, identifiers etc;

(104) Timing, that is the data **260** related to the connection **251**, such as the UE context, is only kept in idle mode for a certain maximum time;

(105) Maintenance actions, for example restarts or manual intervention.

(106) Embodiments of a method for handling the connection **251** will now be described from the viewpoint of the different nodes that participate in embodiments herein. The embodiments will be described in relation to FIG. **5**, FIG. **6**, FIG. **7**, FIG. **8**, FIG. **9**, FIG. **9a**, FIG. **10** and FIG. **10a**. That is, the followings are embodiments related to the first network node **211**, the second network node **212**, the third network node **213** and the wireless device **240**.

(107) Embodiments will first be described from the viewpoint of the first network node **211**. The first network node **211** embodiments relate to FIGS. **5**, **6**, **7** and **7a**.

(108) In some embodiments the fourth network node **214** may take the place of the first network node **211** for some of the actions.

(109) FIGS. **5** and **6** are flow charts that complement FIGS. **3** and **4a-4d** and that describe actions that may take place to resume the connection **251** between the wireless device **240** and the wireless communications network **200** according to embodiments herein. More specifically, FIG. **5** complements FIG. **3** and describes actions to suspend the connection **251** between the wireless device **240** and the wireless communications network **200**, while FIG. **6** complements FIG. **4a** and describes actions to resume the connection **251** from the suspended state.

(110) FIG. **5** further outlines how the wireless device **240** in 'connected mode' is moved to 'idle mode', while keeping the data **260** related the connection **251**, e.g. the UE context, in the wireless device **240** and in the network nodes so that the previous connection **251** may be 'resumed' later. In other words, so that it is possible to continue with the same connection data, e.g. security context, keys, bearer setup etc.

(111) Actions **501**, **502**, **503**, **504**, **505a** and **505b** related to FIG. **5** are performed by the first network node **211**. FIG. **5** also describes two modes or states of operation of the wireless device **240**. These states are referred to as state **500**, in which the wireless device **240** is in RRC Connected Mode, and state **506** in which the wireless device **240** is in a suspended mode, also referred to in FIG. **5** as in Idle Mode, but suspended.

(112) FIG. **6** outlines how the previously suspended wireless device **240**, may resume the previous connection **251**. Parts of the method are performed by the wireless device **240** while other parts are

performed by the first network node **211**. Actions **601-603** are performed by the first network node **211**. Actions **603a-608b** are performed by the first network node **211**.

(113) FIG. **6** also describes two modes or states of operation of the wireless device **240**. These states are referred to as state **600**, in which the wireless device **240** is in a suspended mode, also referred to in FIG. **6** as in Idle Mode, but suspended. In state **609** the wireless device **240** has re-entered in RRC Connected Mode.

(114) FIGS. **7** and **7a** are flowcharts that describe actions of a method for suspending and resuming the connection **251** according to embodiments herein. FIGS. **7** and **7a** describe the same actions but in FIG. **7a** an arrow has been added from start to action **705** compared to FIG. **7**. Actions **701-710** are also related to FIGS. **5** and **6** and are performed by the first network node **211**. FIGS. **5**, **6** and **7** complement each other.

(115) Actions **501, 701**

(116) The first network node **211** may obtain an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** has not been used to transfer data for a while, e.g. that the wireless device **240** has been inactive for a while. In other words the obtained indication is or may be interpreted as an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended or released. The indication may be provided as a timeout of the connection **251** between the wireless device **240** and the wireless communications network **200**. The time out may be provided by a timer module **1110** in the first network node **211**. The timer module **1110** may be implemented by a processor **1180** in the first network node **211**.

(117) Actions **502, 702**

(118) As a result of the obtained indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended or released the first network node may determine whether or not to suspend or release the connection **251**.

(119) In some embodiments the first network node **211** checks that the wireless device **240** supports resumption of the connection **251** before it tries to suspend the connection **251**. This action may be performed by means such as a checking module **1120** in the first network node **211**. The checking module **1120** may be implemented by the processor **1180** in the first network node **211**.

(120) Actions **503, 703**

(121) Other network nodes, e.g. core network nodes such as the second network node **212** and the third network node **213** may need to be informed about the suspension. For example, the network may not send downlink data to the wireless device **240** during the suspension of the connection **251** without paging the wireless device **240** first.

(122) In some embodiments the first network node **211** transmits an indication to the second network node **212**, which is responsible for paging of the wireless device **240**, that the connection **251** between the wireless device **240** and wireless communications network **200** is to be released.

(123) The message to the second network node **212** may comprise a request to deactivate the data **260** related to the connection **251**, e.g. the UE context, in the second network node **212**. The request to deactivate the data **260** related to the connection **251** may further comprises an indication that the second network node **212** shall page the wireless device **240** in case of a downlink control message or a payload is to be transferred to the wireless device **240**. The message to the second network node **212** may further comprise a request to store the UE context, the identifier **270** of the wireless device **240**, e.g. the resume identifier of the wireless device **240** and a description of the bearer.

(124) In some embodiments the identifier **270** of the wireless device **240** is optionally included in the message to the second network node **212** in order to find the data **260** related to the connection **251** in case of larger resume area.

(125) The request may be accepted or acknowledged or it may be rejected by the second network node **212**. This action may be performed by means such as a transmitting module **1130** in the

network node **211**. The transmitting module **1130** may be implemented by a transmitter in the first network node **211**.

(126) This action is related to action **901** below.

(127) Action **504**

(128) The first network node **211** may check whether the second network node **212** accepted the request in actions **503**, **703**.

(129) Actions **505a**, **704**

(130) If the second network node **212** accepted the request in actions **503**, **703** then the first network node **211** may request the third network node **213** to stop DL data traffic to the wireless device **240**, since the wireless device **240** is unable to receive DL data without being paged first during the suspension.

(131) This action may be performed by means such as a requesting module **1140** in the first network node **211**. The requesting module **1140** may be implemented by the processor **1180** in the first network node **211**.

(132) These actions are related to actions **1000** and **1001** below.

(133) Action **705**

(134) In order for the first network node **211** to be able to resume the connection **251** later the first network node **211** stores data **260** related to the connection **251**, and further stores the identifier **270** of the wireless device **240** for locating the stored data **260** related to the connection **251** later.

(135) The data **260** related to the connection **251** may be a UE context related to the connection **251** and the wireless device **240**.

(136) In other words the first network node **211** stores the UE context and the resume identifier of the wireless device **240** and related to the last connection **251**.

(137) This action may be performed by means such as a memory **1190** in the first network node **211**.

(138) Actions **505b**, **706**

(139) The first network node **211** provides the wireless device **240** with a command to transition from a connected mode into a suspended mode. In the suspended mode the connection **251** is suspended and the wireless device **240** stores the data **260** related to the connection **251**. The wireless device **240** further stores the identifier **270** of the wireless device **240** for locating the data **260** related to the connection **251** later.

(140) In other words, the command to transition from the connected mode into the suspended mode further comprises a command to suspend the connection **251** and to store the data **260** related to the connection **251**, and to store the identifier **270** of the wireless device **240** for locating the stored data **260** related to the connection **251** later.

(141) This action may be performed when the first network node **211** has obtained an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended or released. In some embodiments this action is performed if the request to deactivate the UE context has been accepted by the second network node **212**.

(142) These actions may be performed by means such as a commanding module **1150** in the first network node **211**. The commanding module **1150** may be implemented by the processor **1180** in the first network node **211**.

(143) These actions are related to action **801** below.

(144) Actions **603a**, **707**

(145) The first network node **211** receives a message from the wireless device **240**. The received message indicates a request from the wireless device **240** to resume the connection. In other words, the received message indicates that the wireless device **240** wants to establish a new connection **251**.

(146) The received message comprises the identifier **270**, such as the resume identifier, of the wireless device **240** for locating the stored data **260**. The identifier **270** enables the first network

node **211** to locate the data **260** related to the connection.

(147) Thus in some embodiments as part of the receiving action the first network node **211** locates the stored data **260** related to the connection based on the received identifier **270** of the wireless device **240**.

(148) In some embodiments the received message from the wireless device **240** comprises an indication of whether or not to resume data bearers. This was explained above in relation to FIGS. 4-4c. If the data bearers are not resumed at once the first network node **211** and the third network node **213** will perform less processing related to the connection **251** which will enhance the performance of these nodes and of the wireless communications network **200**.

(149) The received message may be received with RRC signaling.

(150) This action may be performed by means such as a receiving module **1160** in the first network node **211**. The receiving module **1160** may be implemented by a receiver in the first network node **211**.

(151) In some embodiments this action may be performed by the fourth network node **214**.

(152) These actions are related to action **803** below.

(153) Action **604**

(154) The first network node **211** may check whether or not it shall accept the access attempt by the wireless device **240**. In some embodiments this action may be performed by the fourth network node **214**.

(155) Action **605a**

(156) If the first network node **211** rejects the random access attempt from the wireless device **240** it may reject the attempt with an indication of when the wireless device **240** earliest may retry to access the network again. The wireless device **240** may try to resume the connection **251** again.

(157) In some embodiments this action may be performed by the fourth network node **214**.

(158) Action **605b**

(159) If the first network node **211** accepts the random access attempt from the wireless device **240** it may further check whether or not the identifier **270** of the wireless device **240** is known and/or valid.

(160) If the identifier **270** is valid then the first network node **211** may check whether or not the related data **260** related to the connection **251** is ok. For example, the 'authentication token' provided by the wireless device **240** may be used to validate that it is the correct wireless device that is trying to resume the connection **251**. The authentication token may be a value derived from a secret key, e.g. by an algorithm, known by the wireless device **240** and by the wireless communications network **200**. This is especially important if the connection **251** was unencrypted.

(161) In some embodiments this action may be performed by the fourth network node **214**.

(162) Action **606a**

(163) If the identifier **270** of the wireless device **240** or the data **260** related to the connection **251** are not valid any more, then the first network node **211** may reject the attempt with an indication that a normal service establishment is needed. This may either be done directly with the assigned scheduling identity or the wireless device **240** may need to make a complete new random access.

(164) In some embodiments this action may be performed by the fourth network node **214**.

(165) Actions **606b**, **708**

(166) The first network node **211** resumes the connection **251** by transmitting a message to the wireless device **240** in response to the message received in action **707** above. The transmitted message comprises the identifier **270** of the wireless device **240**, and further comprises an indication that a resumption of the connection **251** is complete. The suspended connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the data **260** related to the suspended connection **251**. UL payload traffic is now possible.

(167) In other words, the first network node **211** transmits a control message to the wireless device **240** in response to the received message from the wireless device **240**. The control message

comprises the resume identifier of the wireless device **240** and an indication that a resumption of the connection **251** is complete, whereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the UE context.

(168) The control message may further comprise a description of the bearer related to the connection **251**. The bearer may be any one or more out of a signaling bearer and a data bearer. This action may be performed by the transmitting module **1130** in the first network node **211**.

(169) The transmitted message may be transmitted with RRC signaling.

(170) In some embodiments this action may be performed by the fourth network node **214**.

(171) These actions are related to action **804** below.

(172) Actions **606c**, **709**

(173) The first network node **211** may request the second network node **212** to activate the data **260** related to the connection **251**, such as the UE context. In this way the first network node **211** may initiate activation of the data **260** related to the connection **251** in the second network node **212**.

(174) The request may be seen as an indication that the wireless device **240** wants to resume or has resumed the connection **251** again.

(175) This action may be performed by the requesting module **1140** in the first network node **211**.

(176) In some embodiments this action may be performed by the fourth network node **214**.

(177) These actions are related to action **904** below.

(178) Action **607**

(179) In some embodiments the first network node **211** checks whether or not the second network node **212** accepts the request to activate the data **260** related to the connection **251**.

(180) In some embodiments this action may be performed by the fourth network node **214**.

(181) Actions **608a**, **710**

(182) If the second network node **212** accepts the request to activate the UE context the first network node **211** may initiate resumption of bearers for DL in the third network node **213**. In other words, the first network node **211** may transmit an indication that the wireless device **240** wants to establish or has established the connection again. For example, if the RRC Resume complete is sent to the wireless device **240** before the second network node **212** answers, then the indication indicates that the connection has been established. If the first network node **211** waits for the second network node **212** to answer, then the indication indicates that the wireless device **240** wants to establish the connection.

(183) If the second network node **212** does not accept the request to activate the UE context the connection **251** is to be released in the second network node **212** and bearers are to be released in the third network node **213**. Further, the wireless device **240** is to be released and a new normal service request is to be used instead. This may either be done directly with the assigned scheduling identity or the wireless device **240** may need to make a complete new random access.

(184) This action may be performed by means such as an initiating module **1170** in the network node **211**. The initiating module **1170** may be implemented by the processor **1180** in the first network node **211**.

(185) In some embodiments this action may be performed by the fourth network node **214**.

(186) These actions are related to action **1003** below.

(187) Action **608b**

(188) If the second network node **212** rejects the request to activate the UE context the connection is to be released in the second network node **212** and bearers are to be released in the third network node **213**. The wireless device **240** is to be released and a new normal service request used instead. This may either be done directly with the assigned scheduling identity or the wireless device **240** may make a completely new random access.

(189) In some embodiments this action may be performed by the fourth network node **214**.

(190) Embodiments will now be described from the viewpoint of the wireless device **240**. The embodiments related to the wireless device **240** relate to FIGS. **6** and **8**.

(191) FIG. 6 was described above and is a flowchart that describes actions of a method for suspending the connection **251** according to embodiments herein. Actions **601**, **602** and **603b** are performed by the wireless device **240**.

(192) FIG. 8 is a flowchart that describes actions of a method for suspending and resuming the connection **251** according to embodiments herein. Actions **801-804** are related to FIG. 8 and are performed by the wireless device **240**. FIGS. 6 and 8 complement each other.

(193) Action **801**

(194) The wireless device **240** receives from the first network node **211** the command to transition from the connected mode into the suspended mode.

(195) In other words, the wireless device receives the command from the first network node **211** to transition into the suspended idle mode, in which suspended idle mode the wireless device **240** has stored the UE context related to the connection **251**, and the resume identifier of the wireless device **240** for resuming the connection **251** later.

(196) This action may be performed when the first network node **211** has obtained an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended or released. In some embodiments this action is performed if the request to deactivate the UE context has been accepted by the second network node **212**.

(197) The received command may further comprise a command to store a resume area. The resume area indicates in which area, i.e. in which cells, that the resumption is valid.

(198) This action may be performed by means such as a receiving module **1210** in the wireless device **240**. The receiving module **1210** may be implemented by a receiver in the wireless device **240**.

(199) This action is related to actions **504** and **706** above.

(200) Action **802**

(201) The wireless device **240** transitions into the suspended mode, in which suspended mode the connection **251** is suspended and the wireless device **240** stores data **260** related to the connection, and further stores an identifier **270** of the wireless device **240** for locating the stored data **260** related to the connection **251** later.

(202) In other words, the wireless device **240** transitions into the suspended idle mode. This action may be performed by means such as a transitioning module **1220** in the wireless device **240**. The transitioning module **1220** may be implemented by a processor **1280** in the wireless device **240**.

(203) Action **601**

(204) After some time, the wireless device **240** obtains an indication to contact the wireless communications network **200**. Such an indication may be a paging message from the second network node **212**. Thus, the wireless device **240** wants to contact the wireless communications network **200** or in other words the wireless device **240** wants to establish a new connection.

(205) Action **602**

(206) The wireless device **240** may check if it supports suspend and/or resume. It may further check whether the identifier **270** of the wireless device **240** is valid.

(207) Action **603b**

(208) If the wireless device **240** does not support suspend and/or resume or if the identifier **270** of the wireless device **240** is not valid, then the wireless device **240** may do a normal service establishment.

(209) Actions **603a**, **803**

(210) The wireless device **240** transmits a message to the first network node **211**, which transmitted message indicates a request to resume the connection **251** and which message comprises the identifier **270** of the wireless device **240**.

(211) In other words, the wireless device **240** transmits a message to the first network node **211** when the wireless device **240** wants to establish a new connection. The message indicates that the wireless device **240** wants to establish the new connection **251** and comprises the resume identifier

of the wireless device **240**. The message may be transmitted when the wireless device **240** wants to establish the new connection **251**.

(212) The transmitted message to the first network node **211** may be a Random Access RA message.

(213) This action may be performed by a transmitting module **1230** in the wireless device **240**. The transmitting module **1230** may be implemented by a transmitter in the wireless device **240**.

(214) This action is related to actions **603a** and **707** above.

(215) Action **804**

(216) The wireless device **240** receives a message from the first network node **211**, in response to the transmitted message. The received message comprises the identifier **270** of the wireless device **240** and an indication that the resumption of the connection **251** is complete, whereby the suspended connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the data **260** related to the suspended connection **251**.

(217) In other words, the wireless device **240** receives a control message from the first network node **211** in response to the transmitted message. The control message comprises the resume identifier of the wireless device **240** and the indication that the resumption of the connection **251** is complete, whereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the UE context.

(218) The wireless device **240** transitions into connected mode in response to the received message. E.g. the wireless device **240** turns on encryption and in the connected mode the wireless device **240** is prepared to send uplink messages if scheduled, and prepared to receive downlink messages. The previous bearer configuration is active.

(219) This action may be performed by the receiving module **1210** in the wireless device **240**.

(220) This action is related to actions **606a** and **708** above.

(221) Embodiments will now be described from the viewpoint of the second network node **212**. The embodiments related to the second network node **212** relate to FIG. **9**.

(222) FIGS. **9** and **9a** are flowcharts that describe actions of a method for handling, i.e. suspending and resuming, the connection **251** according to embodiments herein. FIG. **9a** is somewhat more detailed than FIG. **9**.

(223) As mentioned above the second network node **212** is responsible for paging the wireless device **240**.

(224) Action **901**

(225) The second network node **212** obtains an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended.

(226) This action may be performed by means such as a receiving module **1310** in the second network node **212**. The receiving module **1310** may be implemented by a receiver in the second network node **212**.

(227) This action is related to actions **502a** and **703** above.

(228) Action **902**

(229) In order for the wireless device **240** to be able to be suspended the second network node **212** may deactivate the data **260** related to the connection **251**. The deactivating of the stored data **260** related to the connection **251** further comprises paging the wireless device **240** in case of a downlink control message or a payload is to be transferred to the wireless device **240**.

(230) The deactivation may be performed in response to an indication that the connection between the wireless device **240** and the wireless communications network **200** is to be suspended.

(231) The deactivating the data **260** related to the connection **251** may further comprise storing the data **260** related to the connection **251**. The deactivating the data **260** related to the connection **251** may further comprises storing a security part of the data **260** related to the connection **251**.

(232) The deactivating the data **260** related to the connection **251** may further comprises storing a S1 signaling identifier.

(233) The deactivating the stored data **260** related to the connection **251** may further comprise commanding the third network node **213** to buffer DL traffic to the wireless device **240**.

(234) This action may be performed by means such as a deactivating module **1320** in the second network node **212**. The deactivating module **1320** may be implemented by a processor **1380** in the second network node **212**.

(235) Action **903**

(236) As a result of the deactivation of the data **260** related to the connection **251** the second network node **212** pages the wireless device **240** in case of a downlink control message is to be transferred to the wireless device **240** or when the third network node **213** has informed the second network node **212** that a payload is to be transferred to the wireless device **240**.

(237) The paging may be performed with a resume identifier. The resume identifier may be obtained from the indication that the connection **251** between the wireless device **240** and wireless communications network **200** is to be suspended.

(238) This action may be performed by means such as a paging module **1330** in the second network node **212**. The paging module **1330** may be implemented by the processor **1380** in the second network node **212**.

(239) Action **904**

(240) When the second network node **212** has obtained the indication that the wireless device **240** wants to or has resumed the connection **251** again, which was described above in relation to actions **606b** and **709**, the second network node **212** activates the stored data **260** related to the connection **251** in response to the indication that the wireless device **240** wants to or has resumed the connection **251** again. Thereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the data related to the connection **251**.

(241) The the second network node **212** may activate the stored data related to the connection **251** by stopping the initiation of paging of the wireless device **240** in case of a downlink control message is to be transferred to the wireless device **240** or in case of a payload is to be transferred to the wireless device **240**. That is, the second network node **212** determines that it shall not page the wireless device **240** anymore.

(242) The activation may further comprise commanding the third network node **213** to stop buffering downlink traffic to the wireless device **240** and to send buffered downlink traffic to the wireless device **240**.

(243) This action may be performed by means such as an activating module **1340** in the second network node **212**. The activating module **1340** may be implemented by the processor **1380** in the second network node **212**.

(244) This action is related to actions **606a** and **709** above and to action **1003** below.

(245) Embodiments will now be described from the viewpoint of the third network node **213**. The embodiments related to the third network node **213** relate to FIG. **10**.

(246) FIGS. **10** and **10a** are flowcharts that describe actions of a method for handling, i.e. suspending and resuming, the connection **251** according to embodiments herein. The method is performed by the third network node **213**. FIG. **10a** complements FIG. **10** and discloses the additional action **1000**.

(247) As mentioned above the third network node **213**, e.g. an SGW, may be responsible for forwarding downlink data packages to the wireless communications device **240**.

(248) Action **1000**

(249) The third network node **213** obtains an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended.

(250) This action is related to actions **505a** and **704** above.

(251) Action **1001**

(252) In order for the wireless device **240** to be able to be suspended the third network node **213** suspends a bearer related to the wireless device **240** in response to the indication that the

connection **251** between the wireless device **240** and the wireless communications network is to be suspended. Suspending the bearer comprises buffering downlink payload to be sent to the wireless device **240** using the bearer.

(253) This action may be performed by means such as a suspending module **1410** in the third network node **213**. The suspending module **1410** may be implemented by a processor **1480** in the third network node **213**.

(254) Action **1002**

(255) When the third network node **213** has received downlink payload to the wireless device **240** for which the bearer is suspended, the third network node **213** informs the second network node **212** that the third network node **213** has received downlink payload to the wireless device **240**.

(256) The indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended, received above in action **1000**, may comprise an indication that the third network node **213** shall inform the second network node **212** that the third network node **213** has received downlink payload to the wireless device **240**.

(257) This action is performed in order to get the second network node **212** to page the wireless device **240**. The paging leads to that the wireless device **240** becomes active again and the downlink data may be transferred.

(258) The bearers in the third network node **213** are originally setup by the second network node **212** and there is a 'connection' between the data **260** related to the connection **251** stored in the second network node **212**, e.g. a UE context in an MME, and the data related to a bearer which is related to the wireless device **240** and stored in the third network node **213**, e.g. a bearer context in an SGW. When the third network node **213** has received the indication of suspension related to the wireless device **240**, this indicates that the second network node **212** must be signalled if downlink data is received in the third network node **213** on the concerned bearers.

(259) This action may be performed by means such as an informing module **1420** in the third network node **213**. The informing module **1420** may be implemented by the processor **1480** and/or a transmitter in the third network node **213**.

(260) This action is related to action **903** above.

(261) Action **1003**

(262) When the third network node **213** receives an indication that the wireless device **240** has resumed the connection **251** again, as described above in actions **608a** and **710** or action **904**, the third network node **213** may transmit the buffered downlink payload to the wireless device **240**. For example, the third network node **213** may have received a command from the second network node **212** to stop buffering downlink traffic to the wireless device **240** and to send buffered downlink traffic to the wireless device **240** in action **904**.

(263) The third network node **213** may transmit the buffered downlink payload in response to the indication that the wireless device **240** has resumed the connection **251** again. Thereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the bearer related to the wireless device **240**, i.e. the bearer used for the last connection **251**.

(264) This action may further comprise stopping the initiation of the buffering of downlink data traffic to the wireless device **240**.

(265) This action may be performed by means such as a transmitting module **1430** in the third network node **213**. The transmitting module **1430** may be implemented by the transmitter in the third network node **213**.

(266) This action is related to actions **608a**, **710** and **904** above.

(267) Embodiments herein may be performed in the first network node **211**. The first network node **211** may comprise the modules mentioned above and depicted in FIG. **11** for handling the connection **251**.

(268) In other words, to perform the method actions for handling the connection **251** between the

wireless device **240** and the wireless communications network **200** described above in relation to FIG. 3-7, the first network node **211** comprises the following arrangement depicted in FIG. 11. (269) Further, embodiments herein may be performed in the system **216** of network nodes **211**, **214** comprising the first network node **211** and the fourth network node **214**. Then the fourth network node **214** may comprise any of the modules comprised in the first network node **211** mentioned above and depicted in FIG. 11 for handling the connection **251**. Particularly, the fourth network node **214** may comprise a receiving module **1560** and a transmitting module **1530** as depicted in FIG. 15. The fourth network node **214** may be configured to perform any of the actions performed by the first network node **211**, such as the actions of receiving and transmitting messages to the wireless device **240**.

(270) The first network node **211** is configured to, e.g. by means of the memory **1190** configured to, store data **260** related to the connection **251**, and further to store an identifier **270** of the wireless device **240** for locating the stored data **260** related to the connection **251** later.

(271) The first network node **211** is further configured to, e.g. by means of the commanding module **1150** configured to, provide the wireless device **240** with the command to transition from the connected mode into the suspended mode. In the suspended mode the connection **251** is suspended and the wireless device **240** stores the data **260** related to the connection **251**, and further stores the identifier **270** of the wireless device **240** for resuming the connection **251** later.

(272) The first network node **211** may further be configured to, e.g. by means of the receiving module **1160** configured to, receive a message from the wireless device **240**. The received message indicates the request from the wireless device **240** to resume the connection. The received message further comprises the identifier **270** of the wireless device **240**. The identifier **270** enables the first network node **211** to identify the data **260** related to the connection **251**.

(273) The first network node **211** may further be configured to, e.g. by means of the transmitting module **1130** configured to, resume the connection **251** by being configured to transmit a message to the wireless device **240**, which transmitted message comprises the identifier **270** of the wireless device **240**. The transmitted message further comprises an indication that the resumption of the connection **251** is complete, whereby the suspended connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the data **260** related to the suspended connection.

(274) The first network node **211** may further be configured to, e.g. by means of an identifying module **1175** configured to, identify the stored data **260** related to the connection based on the received identifier **270** of the wireless device **240**.

(275) The first network node **211** may comprise the memory **1190** comprising one or more memory units. The memory **1190** is configured to store information obtained from for example the wireless communications device **240**, the second network node **212** and the third network node **213**. Such information may be information related to the connection **251**, the wireless device **240**, and bearers related to the wireless device **240**. The data **260** related to the connection **251** and the identifier **270** of the wireless device **240** are examples of such information.

(276) The memory **1190** may also store configurations, schedulings and applications etc. to perform the methods herein when being executed in the first network node **211**.

(277) The embodiments herein for handling the connection **251** between the wireless device **240** and the wireless communications network **200** may be implemented through one or more processors, such as the processor **1180** in the first network node **211** depicted in FIG. 11 together with computer program code for performing the functions and actions of the embodiments herein. The program code mentioned above may also be provided as a computer program product **1191** for instance in the form of a data carrier carrying computer program code **1192** for performing the embodiments herein when being loaded into the first network node **211**. One such carrier may be in the form of a CD ROM disc. It is however feasible with other data carriers such as a memory stick. The computer program code may furthermore be provided as pure program code on a server and

downloaded to the first network node **211**.

(278) Thus, the methods according to the embodiments described herein for the first network node **211** may be implemented by means of a computer program product, comprising instructions, i.e., software code portions, which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the first network node **211**. The computer program product may be stored on a computer-readable storage medium. The computer-readable storage medium, having stored thereon the computer program, may comprise the instructions which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the first network node **211**. In some embodiments, the computer-readable storage medium may be a non-transitory computer-readable storage medium.

(279) Those skilled in the art will also appreciate that the timer module **1110**, the checking module **1120**, the transmitting module **1130**, the requesting module **1140**, the commanding module **1150**, the receiving module **1160**, the initiating module **1170** and the identifying module **1175** described above may refer to a combination of analogue and digital circuits, and/or one or more processors configured with software and/or firmware, e.g. stored in the memory **1190** that when executed by the one or more processors such as the processor **1180** perform as described above. One or more of these processors, as well as the other digital hardware, may be included in a single ASIC, or several processors and various digital hardware may be distributed among several separate components, whether individually packaged or assembled into a SoC.

(280) Embodiments herein may be performed in the wireless device **240**. The wireless device **240** may comprise the modules mentioned above and depicted in FIG. **12** for handling the connection **251**.

(281) In other words, to perform the method actions for handling the connection **251** between the wireless device **240** and the wireless communications network **200** described above in relation to FIG. **3**, FIG. **4a-c**, FIG. **6** and FIG. **8**, the wireless device **240** comprises the following arrangement depicted in FIG. **12**.

(282) The wireless device **220** is configured to, e.g. by means of the receiving module **1210** configured to, receive from the first network node **211** the command to transition from the connected mode into the suspended mode.

(283) The wireless communications device **240** is further configured to, e.g. by means of the transition module **1220** configured to, transition into the suspended mode. In the suspended mode the connection **251** is suspended and the wireless device **240** stores data **260** related to the connection **251**, and further stores an identifier **270** of the wireless device **240** for locating the stored data **260** related to the connection **251** later.

(284) The network node **211** may be further configured to, e.g. by means of the transmitting module **1230** configured to, transmit a message to the first network node **211**. The transmitted message indicates the request to resume the connection **251**. The message comprises the identifier **270** of the wireless device **240**.

(285) The network node **211** may be further configured to, e.g. by means of the receiving module **1210** configured to, receive a message from the first network node **211**, in response to the transmitted message. The received message comprises the identifier **270** of the wireless device **240** and an indication that the resumption of the connection **251** is complete.

(286) The wireless communications device **240** may further comprise a memory **1290** comprising one or more memory units. The memory **1290** is configured to store information obtained from for example the first network node **211**, the second network node **212** and the third network node **213**. Such information may be information related to the connection **251**, the wireless device **240**, and bearers related to the wireless device **240**. The data **260** related to the connection **251** and the identifier **270** of the wireless device **240** are examples of such information.

(287) The memory **1290** may also store configurations, schedulings and applications etc. to

perform the methods herein when being executed in the wireless communications device **240**.
(288) The embodiments herein for handling the connection **251** between the wireless device **240** and the wireless communications network **200** may be implemented through one or more processors, such as the processor **1280** in the wireless device **240** depicted in FIG. **12** together with computer program code for performing the functions and actions of the embodiments herein. The program code mentioned above may also be provided as a computer program product **1291** for instance in the form of a data carrier carrying computer program code **1292** for performing the embodiments herein when being loaded into the wireless device **240**. One such carrier may be in the form of a CD ROM disc. It is however feasible with other data carriers such as a memory stick. The computer program code may furthermore be provided as pure program code on a server and downloaded to the wireless device **240**.

(289) Thus, the methods according to the embodiments described herein for the wireless device **240** may be implemented by means of a computer program product, comprising instructions, i.e., software code portions, which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the wireless device **240**. The computer program product may be stored on a computer-readable storage medium. The computer-readable storage medium, having stored there on the computer program, may comprise the instructions which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the wireless communications device **240**. In some embodiments, the computer-readable storage medium may be a non-transitory computer-readable storage medium.

(290) Those skilled in the art will also appreciate that the receiving module **1210**, the transitioning module **1220** and the transmitting module **1230** described above may refer to a combination of analogue and digital circuits, and/or one or more processors configured with software and/or firmware, e.g. stored in the memory **1290** that when executed by the one or more processors such as the processor **1280** perform as described above. One or more of these processors, as well as the other digital hardware, may be included in a single ASIC, or several processors and various digital hardware may be distributed among several separate components, whether individually packaged or assembled into a SoC.

(291) Embodiments herein may be performed in the second network node **212**. The second network node **212** may comprise the modules mentioned above and depicted in FIG. **13** for handling the connection **251**.

(292) In other words, to perform the method actions for handling the connection **251** between the wireless device **240** and the wireless communications network **200** described above in relation to FIG. **9**, the second network node **212** comprises the following arrangement depicted in FIG. **13**.

(293) The second network node **212** is configured to, e.g. by means of the receiving module **1110** configured to, obtain an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended.

(294) The second network node **212** is further configured to, e.g. by means of the paging module **1130** configured to, page the wireless device **240** in case of a downlink control message is to be transferred to the wireless device **240** or when the third network node **213** has informed the second network node **212** that a payload is to be transferred to the wireless device **240**.

(295) The second network node **212** may further be configured to, e.g. by means of the memory **1390** configured to, store data **260** related to the connection **251**.

(296) The second network node **212** may further be configured to, e.g. by means of the activating module **1140** configured to, activate the stored data **260** related to the connection **251** in response to the indication that the wireless device **240** wants to or has resumed the connection **251** again. Thereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the data **260** related to the connection **251**.

(297) The second network node **212** may be configured to activate the stored data **260** related to the

connection **251** by being configured to stop initiation of paging of the wireless device **240** in case of a downlink control message is to be transferred to the wireless device **240** or in case of a payload is to be transferred to the wireless device **240**.

(298) The second network node **212** may comprise the memory **1390** comprising one or more memory units. The memory **1390** is configured to store information obtained from for example the wireless communications device **240**, the first network node **211** and the third network node **213**. Such information may be information related to the connection **251**, the wireless device **240**, and bearers related to the wireless device **240**. The data **260** related to the connection **251** and the identifier **270** of the wireless device **240** are examples of such information.

(299) The memory **1390** may also store configurations, schedulings and applications etc. to perform the methods herein when being executed in the second network node **212**.

(300) The embodiments herein for handling the connection **251** between the wireless device **240** and the wireless communications network **200** may be implemented through one or more processors, such as the processor **1380** in the second network node **212** depicted in FIG. **13** together with computer program code for performing the functions and actions of the embodiments herein. The program code mentioned above may also be provided as a computer program product **1391** for instance in the form of a data carrier carrying computer program code **1392** for performing the embodiments herein when being loaded into the second network node **212**. One such carrier may be in the form of a CD ROM disc. It is however feasible with other data carriers such as a memory stick. The computer program code may furthermore be provided as pure program code on a server and downloaded to the second network node **212**.

(301) Thus, the methods according to the embodiments described herein for the second network node **212** may be implemented by means of a computer program product, comprising instructions, i.e., software code portions, which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the second network node **212**. The computer program product may be stored on a computer-readable storage medium. The computer-readable storage medium, having stored there on the computer program, may comprise the instructions which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the second network node **212**. In some embodiments, the computer-readable storage medium may be a non-transitory computer-readable storage medium.

(302) Those skilled in the art will also appreciate that the receiving module **1310**, the deactivating module **1320**, the paging module **1330** and the activating module **1340** described above may refer to a combination of analogue and digital circuits, and/or one or more processors configured with software and/or firmware, e.g. stored in the memory **1390** that when executed by the one or more processors such as the processor **1380** perform as described above. One or more of these processors, as well as the other digital hardware, may be included in a single ASIC, or several processors and various digital hardware may be distributed among several separate components, whether individually packaged or assembled into a SoC.

(303) Embodiments herein may be performed in the third network node **213**. The third network node **213** may comprise the modules mentioned above and depicted in FIG. **14** for handling the connection **251**.

(304) In other words, to perform the method actions for handling the connection **251** between the wireless device **240** and the wireless communications network **200** described above in relation to FIG. **10**, the third network node **213** comprises the following arrangement depicted in FIG. **14**.

(305) The third network node **213** is configured to, e.g. by means of an obtaining module **1405** configured to, obtain an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended.

(306) The third network node **213** is further configured to, e.g. by means of the suspending module **1410** configured to, suspend the bearer related to the wireless device **240** in response to the

indication that the connection **251** between the wireless device **240** and the wireless communications network is to be suspended. Suspending the bearer comprises buffering downlink payload to be sent to the wireless device **240** using the bearer.

(307) The third network node **213** may further be configured to, e.g. by means of the informing module **1420** configured to, inform the second network node **212** that the third network node **213** has received downlink payload to the wireless device **240**. This action is performed when the third network node **213** has received downlink payload to the wireless device **240** for which the bearer is suspended

(308) The third network node **213** may further be configured to, e.g. by means of the transmitting module **1430** configured to, transmit the buffered downlink payload to the wireless device **240** in response to an indication that the wireless device **240** has resumed the connection **251** again. Thereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the bearer related to the wireless device **240**.

(309) The third network node **213** may comprise the memory **1490** comprising one or more memory units. The memory **1490** is configured to store information obtained from for example the wireless communications device **240**, the second network node **212** and the first network node **211**. Such information may be information related to the connection **251**, the wireless device **240**, and bearers related to the wireless device **240**. The data **260** related to the connection **251** and the identifier **270** of the wireless device **240** are examples of such information.

(310) The memory **1490** may also store configurations, schedulings and applications etc. to perform the methods herein when being executed in the third network node **213**.

(311) The embodiments herein for handling the connection **251** between the wireless device **240** and the wireless communications network **200** may be implemented through one or more processors, such as the processor **1480** in the third network node **213** depicted in FIG. **14** together with computer program code for performing the functions and actions of the embodiments herein. The program code mentioned above may also be provided as a computer program product **1491** for instance in the form of a data carrier carrying computer program code **1492** for performing the embodiments herein when being loaded into the third network node **213**. One such carrier may be in the form of a CD ROM disc. It is however feasible with other data carriers such as a memory stick. The computer program code may furthermore be provided as pure program code on a server and downloaded to the third network node **213**.

(312) Thus, the methods according to the embodiments described herein for the third network node **213** may be implemented by means of a computer program product, comprising instructions, i.e., software code portions, which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the third network node **213**. The computer program product may be stored on a computer-readable storage medium. The computer-readable storage medium, having stored there on the computer program, may comprise the instructions which, when executed on at least one processor, cause the at least one processor to carry out the actions described herein, as performed by the third network node **213**. In some embodiments, the computer-readable storage medium may be a non-transitory computer-readable storage medium.

(313) Those skilled in the art will also appreciate that the obtaining module **1405**, the suspending module **1410**, the informing module **1420** and the transmitting module **1430** described above may refer to a combination of analogue and digital circuits, and/or one or more processors configured with software and/or firmware, e.g. stored in the memory **1490** that when executed by the one or more processors such as the processor **1480** perform as described above. One or more of these processors, as well as the other digital hardware, may be included in a single ASIC, or several processors and various digital hardware may be distributed among several separate components, whether individually packaged or assembled into a SoC.

(314) As mentioned above embodiments herein provide a more efficient procedure to move the

wireless device **240** from a state corresponding to RRC_IDLE state or mode to a state where data may be transferred to and from the wireless communications network **200** by reusing information from the previous session of the connection **251**.

(315) According to some embodiments herein it is provided a method in the first network node **211** for handling the connection **251** between the wireless device **240** and the wireless communications network **200**. The first network **211** node may:

(316) Obtain an indication that the connection **251** between the wireless device **240** and the wireless communications network is to be suspended or released;

(317) Command the wireless device **240** to transition into a suspended idle mode, in which suspended idle mode the wireless device **240** has stored a UE context related to the connection **251**, and a resume identifier of the wireless device **240** for resuming the connection **251** later;

(318) Receive a first message from the wireless device **240**, which first message indicates that the wireless device **240** wants to establish a new connection and which received message comprises the resume identifier of the wireless device **240**;

(319) Transmit a second message such as e.g. a control message to the UE in response to the received first message from the wireless device **240**. The control message comprises the resume identifier of the wireless device **240** and an indication that a resumption of the connection **251** is complete, whereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the UE context.

(320) According to some other embodiments herein it is provided a method in the wireless device **240** for handling the connection **251** between the wireless device **240** and the wireless communications network **200**. The wireless device **240** may:

(321) Receive a command from the first network node **211** to transition into a suspended idle mode when the first network node **211** has obtained an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be released and/or suspended. In the suspended idle mode the wireless device **240** has stored a UE context related to the connection **251**, and a resume identifier of the wireless device **240** for resuming the connection **251** later.

(322) Transition into the suspended idle mode;

(323) Transmit a first message to the first network node **211** when the wireless device **240** wants to establish a new connection. The message indicates that the wireless device **240** wants to establish the connection **251** and comprises the resume identifier of the wireless device **240**.

(324) Receive a second message such as a control message from the first network node **211** in response to the sent message. The control message comprises the resume identifier of the wireless device **240** and an indication that the resumption of the connection **251** is complete, whereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the UE context.

(325) According to some further embodiments herein it is provided a method in the second network node **212** for handling the connection **251** between the wireless device **240** and the wireless communications network **200**. Then the wireless communications network **200** comprises the second network node **212**. The second network node **212** may:

(326) Deactivate a UE context, which UE context is related to the connection **251**, in response to an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be suspended.

(327) Activate the stored UE context again in response to an indication that the wireless device **240** wants to establish or has established the connection **251** again, whereby the connection **251** between the wireless device **240** and the wireless communications network **200** is resumed using the UE context used for the last connection **251**.

(328) According to some further embodiments herein it is provided a method in a third network node **213** for handling the connection **251** between the wireless device **240** and the wireless

communications network **200**. Then the wireless communications network **200** comprises the third network node **213**. The third network node **213** may:

(329) In response to an indication that the connection **251** between the wireless device **240** and the wireless communications network **200** is to be released/suspended,

(330) suspend a bearer associated with the wireless device **240**. The suspending of the bearer comprises buffering DL data traffic to be sent to the wireless device **240** using the bearer.

(331) Transmit the buffered DL data traffic to the wireless device **240** in response to an indication that the wireless device **240** has established the connection **251** again, whereby the connection **251** between the UE and the wireless communications network is resumed using the bearer used for the last connection **251**.

(332) When using the word “comprise” or “comprising” it shall be interpreted as non-limiting, i.e. meaning “consist at least of”.

(333) Modifications and other embodiments of the disclosed embodiments will come to mind to one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the embodiment(s) is/are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of this disclosure. Although specific terms may be employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

(334) Therefore, the above embodiments should not be taken as limiting the scope, which is defined by the appending claims.

(335) Note that although terminology from 3GPP LTE/SAE has been used in this disclosure to exemplify the embodiments herein, this should not be seen as limiting the scope of the embodiments herein to only the aforementioned system. Other wireless systems may also benefit from exploiting the ideas covered within this disclosure.

(336) Also note that terminology such as a first core network node and a second core network node should be considered to be non-limiting and does in particular not imply a certain hierarchical relation between the two.

Claims

1. A method performed by a first network node for handling a .Iadd.Radio Resource Control (RRC) .Iaddend.connection between a wireless device and a wireless communications network, which wireless communications network comprises the first network node, the method comprising: storing data related to the .Iadd.RRC .Iaddend.connection, and further storing an identifier of the wireless device for locating the stored data related to the .Iadd.RRC .Iaddend.connection later.Iadd., the identifier of the wireless device comprising a radio network temporary identifier (RNTI).Iaddend.; providing the wireless device with a command to transition from a connected mode into a suspended mode, .Iadd.the command comprising the identifier of the wireless device, and .Iaddend.in which suspended mode the .Iadd.RRC .Iaddend.connection is suspended and the wireless device stores the data related to the .Iadd.RRC .Iaddend.connection, and further stores the identifier of the wireless device for locating the stored data later when resuming the .Iadd.RRC .Iaddend.connection; receiving a message from the wireless device, which message indicates a request from the wireless device to resume the .Iadd.RRC .Iaddend.connection and which received message comprises the identifier of the wireless device for locating the stored data, wherein the identifier enables the first network node to locate the data related to the .Iadd.RRC .Iaddend.connection; and resuming the .Iadd.RRC .Iaddend.connection by transmitting a message to the wireless device, which transmitted message comprises .Iadd.the identifier of the wireless device, and further comprises.Iaddend. an indication that a resumption of the .Iadd.RRC .Iaddend.connection is complete, whereby the suspended .Iadd.RRC .Iaddend.connection between the wireless device and

the wireless communications network is resumed using the data related to the suspended .Iadd.RRC .Iaddend.connection.

2. The method according to claim 1, wherein the receiving further comprises: locating the stored data related to the .Iadd.RRC .Iaddend.connection based on the received identifier of the wireless device.

3. The method according to claim 1, wherein the data related to the .Iadd.RRC .Iaddend.connection is a UE context related to the .Iadd.RRC .Iaddend.connection and the wireless device.

4. The method according to claim 1, wherein the message received from the wireless device comprises an indication of whether or not to resume data bearers.

5. The method according to claim 1, wherein the wireless communications network is a Long Term Evolution, LTE, network; the first network node is an evolved NodeB; the wireless device is a User Equipment, UE; [. the connection is a Radio Resource Control, RRC, connection;.] and messages between the first network node and the wireless device are transmitted with RRC signaling.

6. A first network node for handling a .Iadd.Radio Resource Control (RRC) .Iaddend.connection between a wireless device and a wireless communications network, which wireless communications network comprises the first network node, the first network node is configured to: store data related to the .Iadd.RRC .Iaddend.connection, and further to store an identifier of the wireless device for locating the stored data related to the .Iadd.RRC .Iaddend.connection later.Iadd., the identifier of the wireless device comprising a radio network temporary identifier (RNTI).Iaddend.; provide the wireless device with a command to transition from a connected mode into a suspended mode, .Iadd.the command comprising the identifier of the wireless device, and .Iaddend.in which suspended mode the .Iadd.RRC .Iaddend.connection is suspended and the wireless device stores the data related to the .Iadd.RRC .Iaddend.connection, and further stores the identifier of the wireless device for locating the stored data later when resuming the .Iadd.RRC .Iaddend.connection; receive a message from the wireless device, which received message indicates a request from the wireless device to resume the .Iadd.RRC .Iaddend.connection and which received message comprises the identifier of the wireless device, wherein the identifier enables the first network node to identify the data related to the .Iadd.RRC .Iaddend.connection; and resume the .Iadd.RRC .Iaddend.connection by being configured to transmit a message to the wireless device, which transmitted message comprises [.the identifier of the wireless device, and further comprises.] an indication that a resumption of the .Iadd.RRC .Iaddend.connection is complete, whereby the suspended .Iadd.RRC .Iaddend.connection between the wireless device and the wireless communications network is resumed using the data related to the suspended .Iadd.RRC .Iaddend.connection.

7. The first network node according to claim 6, further configured to: identify the stored data related to the .Iadd.RRC .Iaddend.connection based on the received identifier of the wireless device.

8. The first network node according to claim 6, wherein the data related to the .Iadd.RRC .Iaddend.connection is a UE context related to the .Iadd.RRC .Iaddend.connection and the wireless device.

9. The first network node according to claim 6, wherein the message received from the wireless device comprises an indication of whether or not to resume data bearers.

10. The first network node according to claim 6, wherein the wireless communications network is a Long Term Evolution, LTE, network; the first network node is an evolved NodeB; the wireless device is a User Equipment, UE; [. the connection is a Radio Resource Control, RRC, connection;.] and messages between the first network node and the wireless device are transmitted with RRC signaling.

11. A method performed by a wireless device for handling a .Iadd.Radio Resource Control (RRC) .Iaddend.connection between the wireless device and a wireless communications network, which wireless communications network comprises a first network node, the method comprising:

receiving from the first network node a command to transition from a connected mode into a suspended mode.Iadd., the command comprising an identifier of the wireless device, the identifier of the wireless device comprising a radio network temporary identifier (RNTI).Iaddend.; transitioning into the suspended mode, in which suspended mode the .Iadd.RRC .Iaddend.connection is suspended and the wireless device stores data related to the .Iadd.RRC .Iaddend.connection, and further stores .[.an.]. .Iadd.the .Iaddend.identifier of the wireless device for locating the stored data related to the .Iadd.RRC .Iaddend.connection later when resuming the .Iadd.RRC .Iaddend.connection; transmitting a message to the first network node, which transmitted message indicates a request to resume the .Iadd.RRC .Iaddend.connection and which transmitted message comprises the identifier of the wireless device; and receiving a message from the first network node, in response to the transmitted message, which received message comprises .[.the identifier of the wireless device and.]. an indication that a resumption of the .Iadd.RRC .Iaddend.connection is complete, whereby the suspended .Iadd. RRC .Iaddend.connection between the wireless device and the wireless communications network is resumed using the data related to the suspended .Iadd.RRC .Iaddend.connection.

12. The method according to claim 11, wherein the data related to the .Iadd.RRC .Iaddend.connection is a UE context related to the .Iadd.RRC .Iaddend.connection and the wireless device.

13. The method according to claim 11, wherein the transmitted message comprises an indication of whether to resume data bearers.

14. The method according to claim 11, wherein the wireless communications network is an Long Term Evolution, LTE, network; the first network node is an evolved NodeB; the wireless device is a User Equipment, UE; .[. the connection is a Radio Resource Control, RRC, connection;.]. and messages between the first network node and the wireless device are transmitted with RRC signaling.

15. A wireless device for handling a .Iadd.Radio Resource Control (RRC) .Iaddend.connection between the wireless device and a wireless communications network, which wireless communications network comprises a first network node, the wireless device is configured to: receive from the first network node a command to transition from a connected mode into a suspended mode.Iadd., the command comprising an identifier of the wireless device, the identifier of the wireless device comprising a radio network temporary identifier (RNTI).Iaddend.; transition into the suspended mode, in which suspended mode the .Iadd.RRC .Iaddend.connection is suspended and the wireless device stores data related to the .Iadd.RRC .Iaddend.connection, and further stores .[.an.]. .Iadd.the .Iaddend.identifier of the wireless device for locating the stored data related to the .Iadd.RRC .Iaddend.connection later when resuming the .Iadd.RRC .Iaddend.connection; transmit a message to the first network node, which transmitted message indicates a request to resume the .Iadd.RRC .Iaddend.connection and which message comprises the identifier of the wireless device; and receive a message from the first network node, in response to the transmitted message, which received message comprises .[.the identifier of the wireless device and.]. an indication that a resumption of the .Iadd.RRC .Iaddend.connection is complete, whereby the suspended .Iadd.RRC .Iaddend.connection between the wireless device and the wireless communications network is resumed using the data related to the .Iadd.RRC .Iaddend.suspended connection.

16. The wireless device according to claim 15, wherein the data related to the .Iadd.RRC .Iaddend.connection is a UE context related to the .Iadd.RRC .Iaddend.connection and the wireless device.

17. The wireless device according to claim 15, wherein the transmitted message comprises an indication of whether to resume data bearers.

18. The wireless device according to claim 15, wherein the wireless communications network is an Long Term Evolution, LTE, network; the first network node is an evolved NodeB; the wireless

device is a User Equipment, UE; .[. the connection is a Radio Resource Control, RRC, connection;.] and messages between the first network node and the wireless device are transmitted with RRC signaling.
